Panel Data Replication Report

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Reference and data

APA reference. Huntington and Liddle (2022). How energy prices shape OECD economic growth: Panel evidence. *Energy Economics*, 111, 106082. https://doi.org/10.1016/j.eneco.2022.106082

Link to the paper. https://doi.org/10.1016/j.eneco.2022.106082

Data used. The replication uses the author-provided dataset growth$_{E}E.xlsx(includedinthisrepository).T$

Formatted datasets. The same dataset was prepared for Python, Stata, and R workflows. The Python pipeline is in replication$_{f}inal.py.TheStataandRtranslationsarein$replication.do$and$re

LLM translation difficulties (R and Stata).

- No official R package provides xtdcce2; CCEMG must be coded manually.
- Pesaran CIPS and delta tests are not in plm and require manual or external implementati
- Panel IV estimation per country is not native in plm; ivreg loops are needed.
- Panel IRFs and some robustness steps require manual post-estimation code.

Abstract (about 100 words)

This report partially replicates Huntington and Liddle (2022) on the effect of energy prices on OECD growth. Using the author-provided dataset for 18 OECD countries over 1960--2016 (with a baseline sample from 1972), I reproduce the main descriptive statistics, unit-root tests, cross-section dependence tests, and the core CCE-MG estimates with robustness checks. I also estimate static panel models (between, within, Mundlak, TWFE, first differences) and dynamic ARDL specifications with Anderson--Hs IV. The results confirm a negative short-run energy price effect and show heterogeneous responses across countries.

Introduction (summary of new insights)

Three replication-based insights stand out. First, energy-price effects on growth are heterogeneous across countries, and this heterogeneity is visible even before imposing a common slope model. Second, first-difference stationarity is strong for all core variables, while levels show weaker evidence, reinforcing the use of differenced specifications. Third, robustness checks (Great Moderation split, recession dummy, country exclusions) yield similar qualitative results, suggesting that the negative short-run effect of energy prices is not driven by single episodes or outliers.

Table 1: Key correlation of the paper

Dependent variable (Y). Real GDP (log level) and its first difference (growth).

Key explanatory variable (X). Energy prices (log level) and its first difference.

Level. Country-year panel (OECD countries).

Links to data. This replication uses the author-supplied dataset; source links are documented in the paper and in the Excel workbook metadata.

Database update

No update to post-2016 data was attempted in this replication. The dataset remains as provided by the authors. If updates are added later, the same pipeline can be rerun to regenerate all outputs.

Panel data sample selection

The full dataset includes 18 countries with a maximum span from 1960 to 2016. All countries have at least three consecutive observations of the dependent variable, so no unit is excluded for the dynamic panel requirement. The panel is slightly unbalanced due to a shorter time span for Australia and Germany.

Individuals excluded: None.

Panel size summary: 18 countries; unbalanced; maximum $T = 57$ for most countries.

Number of individuals per date

yr	n$_{o}bs$
1960	16
1961	16
1962	17
1963	17
1964	17
1965	17
1966	17
1967	17
1968	17
1969	17
1970	17
1971	17
1972	18
1973	18
1974	18
1975	18
1976	18
1977	18
1978	18
1979	18
1980	18
1981	18
1982	18
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2001	18
2002	18
2003	18
2004	18
2005	18
2006	18
2007	18
2008	18
2009	18
2010	18
2011	18
2012	18
2013	18

Number of individuals by their time observations

Observations per country	Number of countries
57	16
55	1
45	1

Panel data variable classification

The within-variance share is reported below. No variable is fully time-invariant in this dataset. The ordering shows that CPI and energy prices have high within variation, while government expenditure and trade openness have lower within shares.

variable	N	NT	NT$_{o}ver_{N}$	var$_{t}otal$	var$_{b}etween$	var
lcpi	18	1012	56.22222222222222	0.838120589189309	0.04403253979767908	0.7
lenpr	18	1012	56.22222222222222	0.06976723028426418	0.016064665172727734	0.0
lrgdpmad	18	1012	56.22222222222222	0.1860727512696083	0.043235948026990496	0.1
iy	18	1012	56.22222222222222	0.001489036294307232	0.0005666330295944054	0.0
open	18	1012	56.22222222222222	0.08647569736853698	0.0640777522788165	0.0
expgdp	18	1012	56.22222222222222	0.008060373061907957	0.006198108209561397	0.0

Descriptive statistics and tests

Table 1: Data summary

Variable	Mean	Standard deviation	Coeff. of variation	Minimum	Maximum
GDP	9.637	0.431	0.045	7.992	10.520
CPI	4.189	0.915	0.219	0.978	5.394
ENERGY	4.335	0.264	0.061	3.467	4.891
OpenTrade	0.518	0.294	0.568	0.089	2.151
GovExp	0.249	0.090	0.360	0.000	0.475
Invest	0.233	0.039	0.166	0.146	0.389
D.GDP	0.024	0.025	1.071	-0.090	0.214
D.CPI	0.045	0.042	0.946	-0.046	0.321
D.ENERGY	0.006	0.071	11.101	-0.242	0.423
D.OpenTrade	0.005	0.082	16.693	-1.057	1.016
D.GovExp	0.001	0.019	19.538	-0.334	0.120
D.Invest	-0.000	0.014	-93.573	-0.082	0.117

Table 2: CIPS unit-root tests

Variable	Lag 0	Lag 1	Lag 2	Lag 3
GDP	-1.46	-1.55	-1.31	-1.39
D.GDP	-5.68**	-4.87**	-4.33**	-4.07**
CPI	-1.85	-2.57*	-1.80	-1.65
D.CPI	-4.03**	-4.15**	-3.96**	-3.66**
ENERGY	-2.05*	-1.91	-1.80	-1.68
D.ENERGY	-8.01**	-6.49**	-5.87**	-5.33**
OpenTrade	-2.17*	-2.27*	-2.04*	-1.85
D.OpenTrade	-6.52**	-5.76**	-5.13**	-3.96**
GovExp	-1.86	-1.76	-1.77	-1.84
D.GovExp	-7.54**	-5.16**	-4.38**	-3.99**
Invest	-1.99*	-2.27*	-2.14*	-1.98*
D.Invest	-6.03**	-5.11**	-4.49**	-4.09**

Cross-section dependence tests (Pesaran CD)

variable	cd$_{s}tat$	pvalue	avg$_{c}orr$
lrgdpmad	91.34667847594903	0.0	0.9781593644692392
lcpi	92.5044919665393	0.0	0.9905574738151404
lenpr	72.21264561782723	0.0	0.7732681332553775
open	62.292797676985444	0.0	0.6670443239244221
expgdp	41.19491319569362	0.0	0.4411237582270121
iy	17.064109818798798	0.0	0.18272606179089781

Slope heterogeneity (Pesaran–Yamagata delta test)

delta$_{s}tat$	pvalue
16682.986343800585	0.0

Static panel estimators

The following table reports between, within, Mundlak (random effects with means), two-way fixed effects, and first-difference estimators.

model	variable	coef	se	t
between	Intercept	7.367149430197046	1.5585992442371168	4.7267759543942445
between	lcpi	0.8352204929003486	0.15407162295576393	5.420988478456879
between	lenpr	-0.25460085340916955	0.2954822181578972	-0.8616452624337555
between	open	-0.23340230182844357	0.13136776990567015	-1.7767090207593557
between	expgdp	0.527617705811679	0.5217586198101073	1.0112294953626335
between	iy	-0.5857030085032255	1.434780492601186	-0.4082178504123477
within	Intercept	7.894913329329997	0.4134363890366767	19.095835631995193
within	lcpi	0.3804465894315625	0.03780587053580833	10.063161726992043
within	lenpr	0.024843501044482897	0.08004879931269464	0.3103544495081897
within	open	0.33760250440516903	0.06443609559361241	5.239338313332501
within	expgdp	-0.7063389052122699	0.3064628461764012	-2.3048108898841804
within	iy	0.18197270941175334	0.7450297819401994	0.24424890631601565
re$_{m}undlak$	Intercept	7.366026385909653	2.0309580271124963	3.6268727800260168
re$_{m}undlak$	lcpi	0.3804465894315623	0.008306282511075031	45.8022694176728
re$_{m}undlak$	lenpr	0.024843501044482928	0.025210601891649666	0.985438632177666
re$_{m}undlak$	open	0.3376025044051691	0.03288056988623851	10.267538110599041
re$_{m}undlak$	expgdp	-0.7063389052122689	0.122317854460577	-5.774618172688124
re$_{m}undlak$	iy	0.18197270941175092	0.15363908180732083	1.1844167985849028
re$_{m}undlak$	lcpi$_{m}ean$	0.45493791520200094	0.20093854948542447	2.2640648913164414
re$_{m}undlak$	lenpr$_{m}ean$	-0.2793647195801001	0.38577819009066167	-0.7241589254033429
re$_{m}undlak$	open$_{m}ean$	-0.5710715221692612	0.17427621819576503	-3.2768184212476696
re$_{m}undlak$	expgdp$_{m}ean$	1.234176176797533	0.6907551601220036	1.786705692621317
re$_{m}undlak$	iy$_{m}ean$	-0.7672943782658251	1.875323284329246	-0.4091531229188925
twfe	Intercept	9.133827528832063	0.7654687304246418	11.932332655528718
twfe	lcpi	0.13607373525045122	0.04350987239654234	3.1274220712553684
twfe	lenpr	-0.07024801331251716	0.131582383758702	-0.5338709582989404
twfe	open	0.15899758365735372	0.089787973269391	1.7708115894354026
twfe	expgdp	0.136060017024902	0.3478716137285986	0.39112135527980413
twfe	iy	0.5231171488499312	0.4495262947196942	1.163707562815931
fd	lcpi	0.5129529432274016	0.006053054737666643	84.74282250174015
fd	lenpr	-0.09833348377596844	0.021417512530656657	-4.591265378517491
fd	open	0.0899564064272665	0.01856893837572467	4.844456080745427
fd	expgdp	-0.4917006103619636	0.08097613760554283	-6.0721667506194645
fd	iy	0.6795021832321125	0.10918257635539921	6.223540476094566

CCE-MG estimates and robustness

Table 3: CCE-MG baseline and robust estimates

Variable	cce	ccerobust	dccecce
D.CPI	-0.131*	-0.179**	-0.127**
	(0.063)	(0.055)	(0.048)
D.Energy	-0.022	-0.024*	-0.004
	(0.014)	(0.012)	(0.012)
D.OpenTrade	0.219**	0.202**	0.156**
	(0.046)	(0.041)	(0.035)
D.GovExp	-0.296**	-0.193**	-0.173*
	(0.082)	(0.059)	(0.075)
D.Invest	0.653**	0.653**	0.571**
	(0.069)	(0.067)	(0.077)
L.GDP	-0.353**	-0.349**	-0.338**
	(0.044)	(0.039)	(0.044)
L.CPI	-0.049	-0.051	-0.091**
	(0.029)	(0.027)	(0.028)
L.OpenTrade	0.088**	0.058**	0.136**
	(0.029)	(0.017)	(0.030)
L.Invest	0.273**	0.261**	0.294**
	(0.069)	(0.063)	(0.061)
$_{c}ons$	0.228	0.186	0.113
	(0.240)	(0.223)	(0.213)

Table 4: CCE-MG robustness checks

Variable	inst	exog	w/o$_{m}od$	recession	ger$_{i}re$	sixties
	(0.068)	(0.068)	(0.064)	(0.070)	(0.068)	(0.040)
D.OpenTrade	0.088	0.087	0.082	0.098*	0.088	0.144**
	(0.046)	(0.048)	(0.043)	(0.047)	(0.046)	(0.038)
D.GovExp	-0.093	-0.104	-0.115	-0.114	-0.093	-0.113
	(0.076)	(0.074)	(0.075)	(0.064)	(0.076)	(0.073)
D.Invest	0.514**	0.538**	0.580**	0.541**	0.514**	0.555**
	(0.135)	(0.129)	(0.129)	(0.131)	(0.135)	(0.098)
L.GDP	-0.431**	-0.474**	-0.445**	-0.444**	-0.431**	-0.350**
	(0.079)	(0.076)	(0.075)	(0.082)	(0.079)	(0.048)
L.CPI	-0.109*	-0.090	-0.075	-0.107	-0.109*	-0.078*
	(0.054)	(0.051)	(0.059)	(0.056)	(0.054)	(0.032)
L.OpenTrade	0.127**	0.146**	0.135**	0.129**	0.127**	0.149**
	(0.039)	(0.041)	(0.041)	(0.042)	(0.039)	(0.037)
L.Invest	0.464**	0.491**	0.491**	0.487**	0.464**	0.322**
	(0.134)	(0.124)	(0.129)	(0.138)	(0.134)	(0.080)
mod	-0.000	-0.001		-0.001	-0.000	-0.000
	(0.007)	(0.007)		(0.007)	(0.007)	(0.006)
D.Energy×mod	0.024	-0.001		0.030	0.024	-0.001
	(0.032)	(0.026)		(0.030)	(0.032)	(0.024)
D.Energy	-0.024		-0.006	-0.025	-0.024	-0.017
	(0.027)		(0.016)	(0.023)	(0.027)	(0.025)
recess				0.005		
				(0.005)		
$_{c}ons$	-0.331	-0.043	-0.151	-0.268	-0.331	-0.062
	(0.812)	(0.867)	(0.804)	(0.826)	(0.812)	(0.252)
Price effect (post-1982)	0.000		-0.006	0.005	0.000	-0.018
Observations	808	808	808	808	808	976

Table 5: Country responses and energy intensity

Country	Intensity	Exports	Imports	$\text{Post}_1 982$	Pre-1983
Australia	6.617	9.462	4.660	-0.041	0.045
Belgium	6.230	0.490	2.477	-0.063	-0.063
Canada	10.861	16.607	12.377	0.042	-0.024
Switzerland	3.072	0.605	1.230	0.021	-0.085
Germany	4.224	5.535	14.295	-0.142	-0.314
Denmark	3.614	0.738	0.817	0.012	-0.092
Spain	3.996	1.488	5.077	0.032	0.017
Finland	6.438	0.419	1.184	-0.097	-0.221
France	4.697	4.758	10.227	-0.003	-0.090
United Kingdom	4.615	8.877	9.302	0.038	0.040
Ireland	3.274	0.074	0.506	0.067	0.012
Italy	3.390	1.311	7.116	0.006	-0.092
Japan	4.542	3.618	20.419	-0.091	-0.221
Netherlands	5.769	2.729	3.772	-0.009	-0.069
Norway	7.840	8.174	1.790	-0.095	-0.123
Portugal	3.369	0.154	0.903	0.248	0.017
Sweden	6.607	1.392	2.232	-0.124	-0.148
United States	7.221	72.613	91.547	-0.073	-0.140
Average				-0.015	-0.086

Table 6: Country responses and intensity regressions

Specification	Coef	t-stat	N	F	RMSE
(1) pre-1983/1982	-0.002	-0.20	18	0.04	0.096
(2) post-1983/1982	-0.015	-1.06	18	1.12	0.083
(3) pre-1983/1982 (excl. outliers)	-0.008	-0.87	16	0.76	0.072
(4) post-1983/1982 (excl. outliers)	-0.010	-0.89	16	0.79	0.054

Dynamic ARDL and Anderson–Hsiao IV

ARDL (OLS)

	coef	se	t	p
const	0.015738000689406267	0.0011265847327601517	13.969655572065047	2.387786134
$\text{l}_d lrgdpmad$	0.3119639106312253	0.0389638636373072	8.00649323524799	1.180256375
dlenpr	-0.03881579655145945	0.012385393685738703	-3.1339977990489167	0.001724421
$\text{l}_d lenpr$	-0.040394086246393755	0.010785047520982536	-3.745378605685901	0.000180121
dlcpi2	0.015545092992315608	0.018374193970997335	0.8460285668504801	0.397536808
dopen	0.059600278074169885	0.028387414589408393	2.099531744479728	0.035770052
dexpgdp	-0.14991259782368957	0.09573382391571318	-1.5659313677439328	0.117364685
diy	0.5519649943857072	0.07924292282416787	6.9654800039425915	3.272859088
$\text{l}_e cterm$	-0.015671026700346966	0.003564226938142924	-4.396753341556885	1.098820924

ARDL (IV)

	coef	se	t	p
const	0.002671172387805473	0.002370789245892842	1.1267017481343038	0.259868603
$l_{d}lenpr$	-0.013438166466700168	0.01568398964927212	-0.8568079147721077	0.391551050
dlcpi2	-0.08491373492609669	0.04622790286539173	-1.8368502498015522	0.066232005
dopen	0.07361721046625029	0.033116187948566043	2.2229977248766635	0.026215953
dexpgdp	-0.06337736099761855	0.1495860738780156	-0.4236849016393164	0.671795627
diy	0.14516498079718212	0.10307938539728925	1.4082833365535339	0.159047187
$l_{e}cterm$	-0.013315186686480861	0.004448367273750713	-2.9932750303807416	0.002760009
$l_{d}lrgdpmad$	1.0156049102559086	0.12968571184439773	7.831278371471433	4.884981308
dlenpr	-0.008641587347971935	0.05467493455834732	-0.15805391296353383	0.874414316

First-stage R2 (IV)

endog	r2

Hausman test (OLS vs IV)

$hausman_{s}tat$	pvalue
53.54924961784471	2.2993157600126324e-08

Impulse response (energy price shock)

h	tau	se	lower	upper
1	-0.008641587347971935	0.05467493455834732	-0.11580445908233268	0.09852128438638881
2	-0.0222146050097058	0.057711497314770835	-0.13532913974665664	0.09089992972724503
3	-0.0362091297756697	0.060806251891808945	-0.15538938348361525	0.08297112393227583
4	-0.05042203784468056	0.06394897258134417	-0.17576202410411512	0.07491794841475402

Long-run effect

$lt_{c}oef$	rho
	1.0156049102559086

Figures

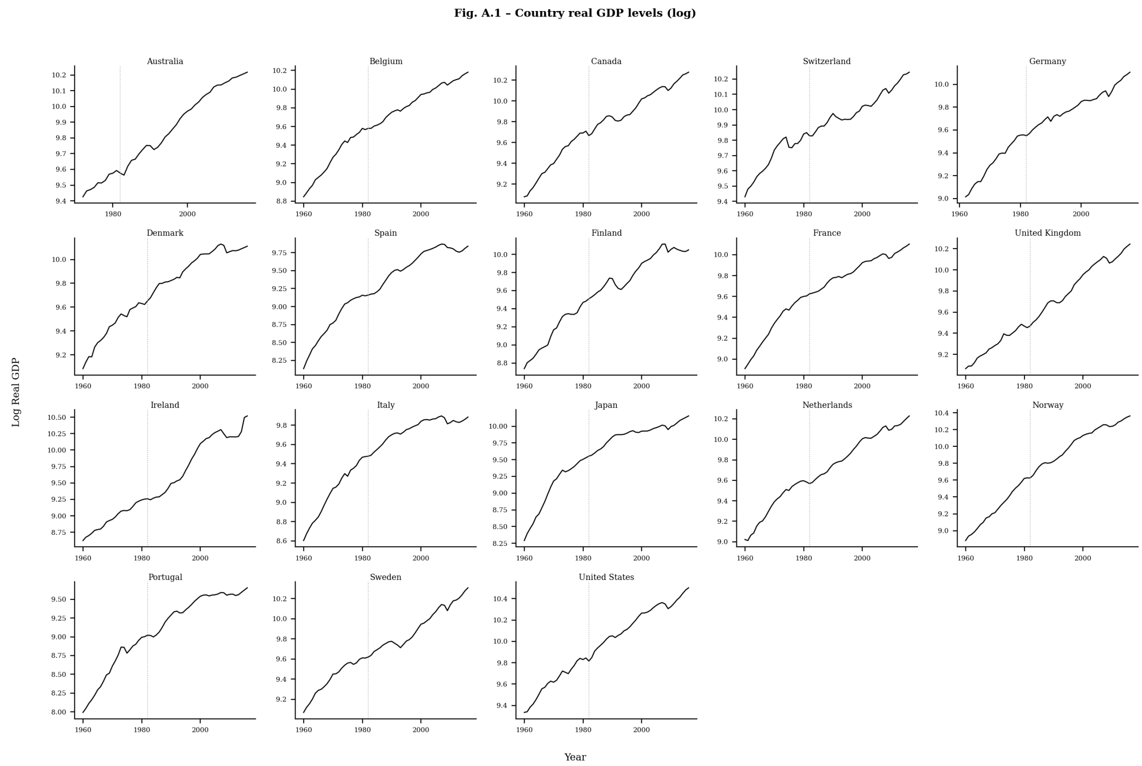

Figure 1: Fig. A1 – GDP levels (log)

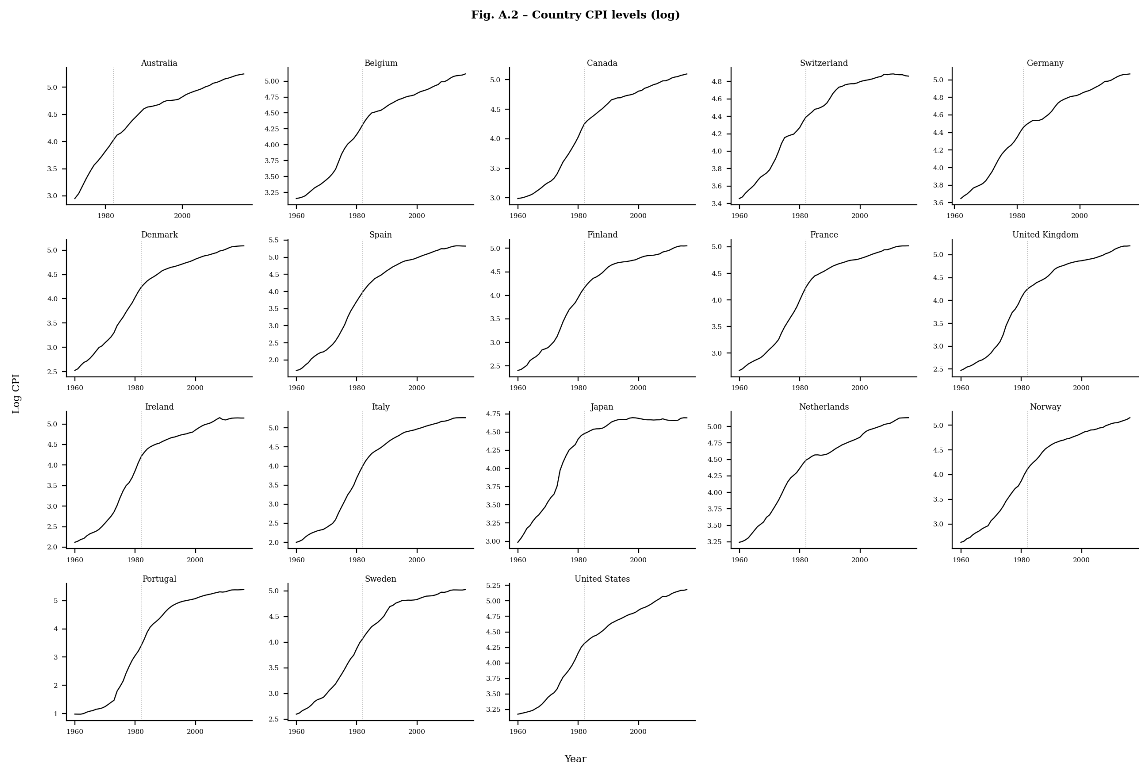

Figure 2: Fig. A2 – CPI levels (log)

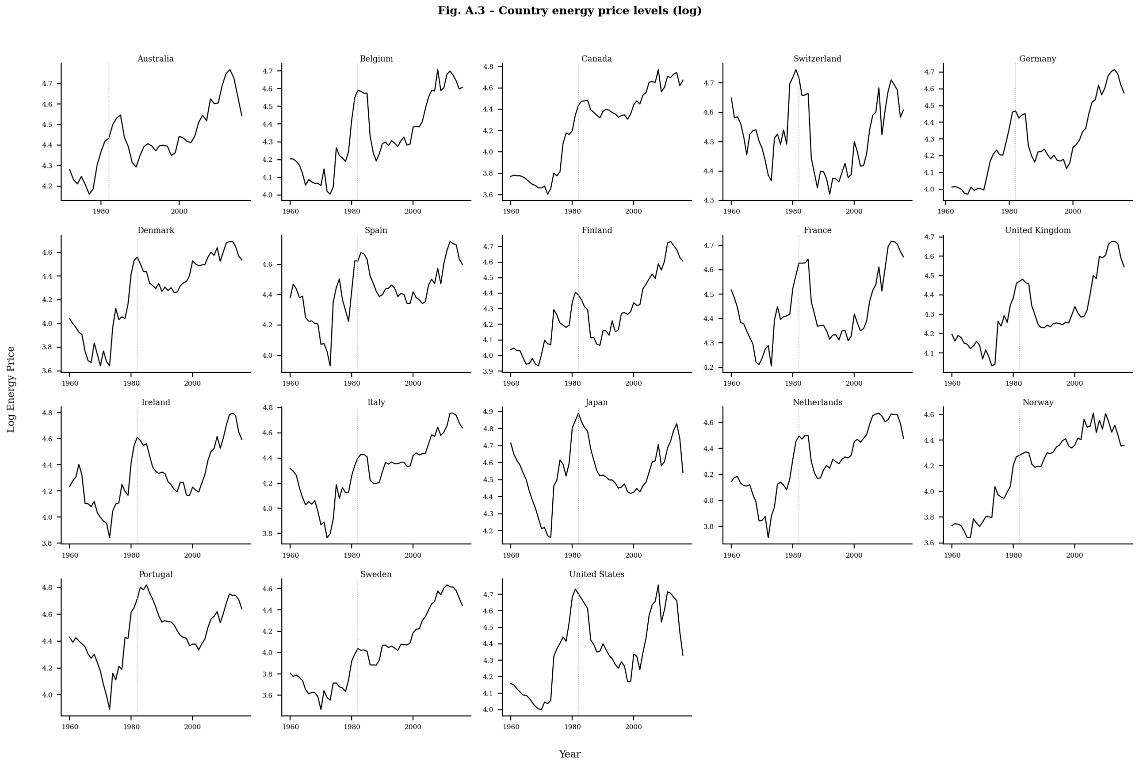

Figure 3: Fig. A3 – Energy price levels (log)

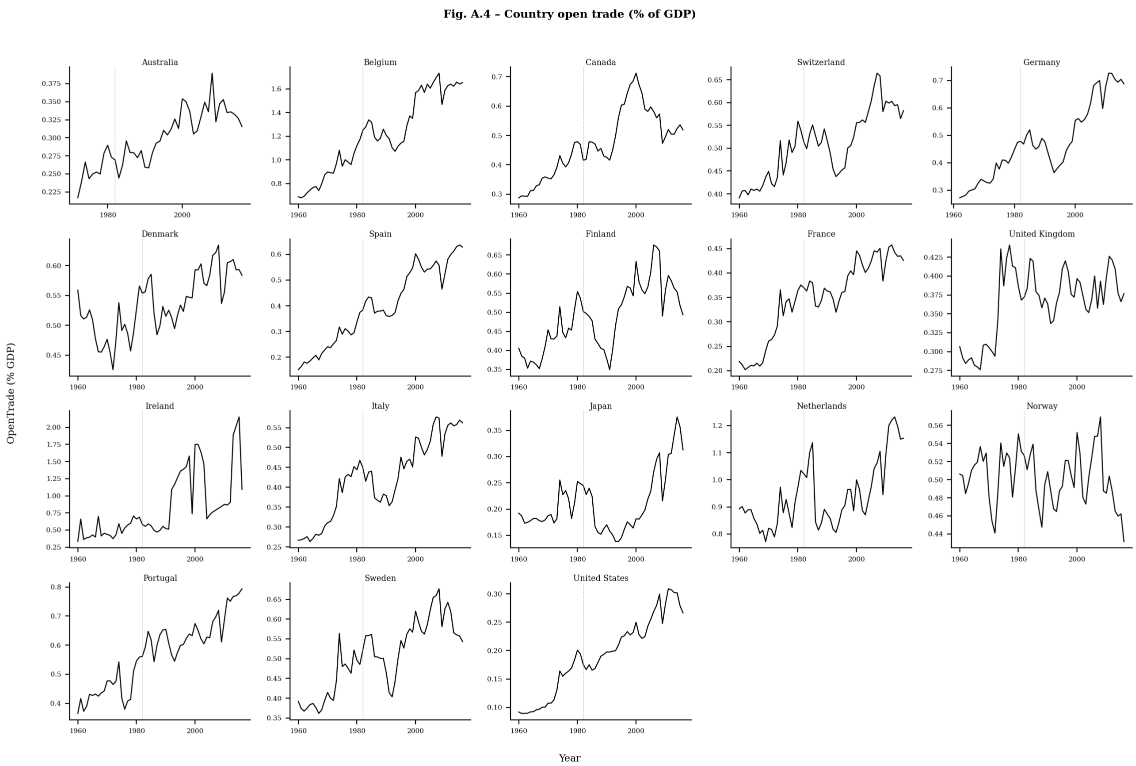

Figure 4: Fig. A4 – Open trade ratio

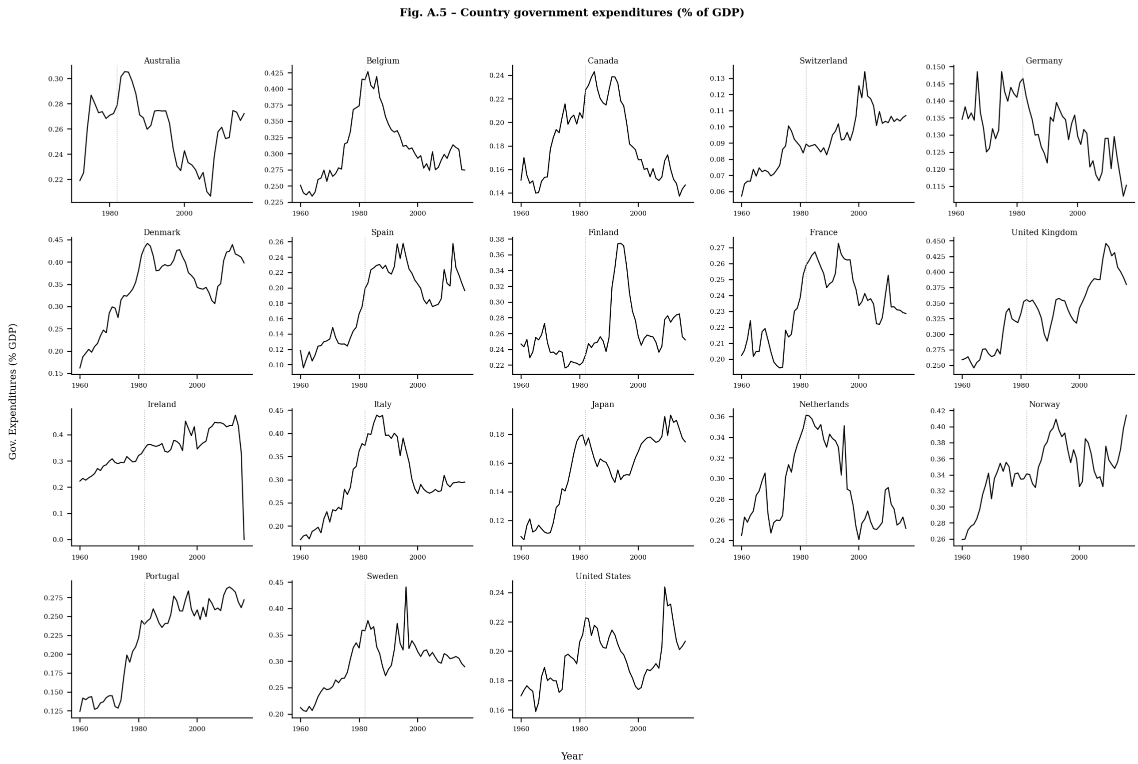

Figure 5: Fig. A5 – Government expenditure ratio

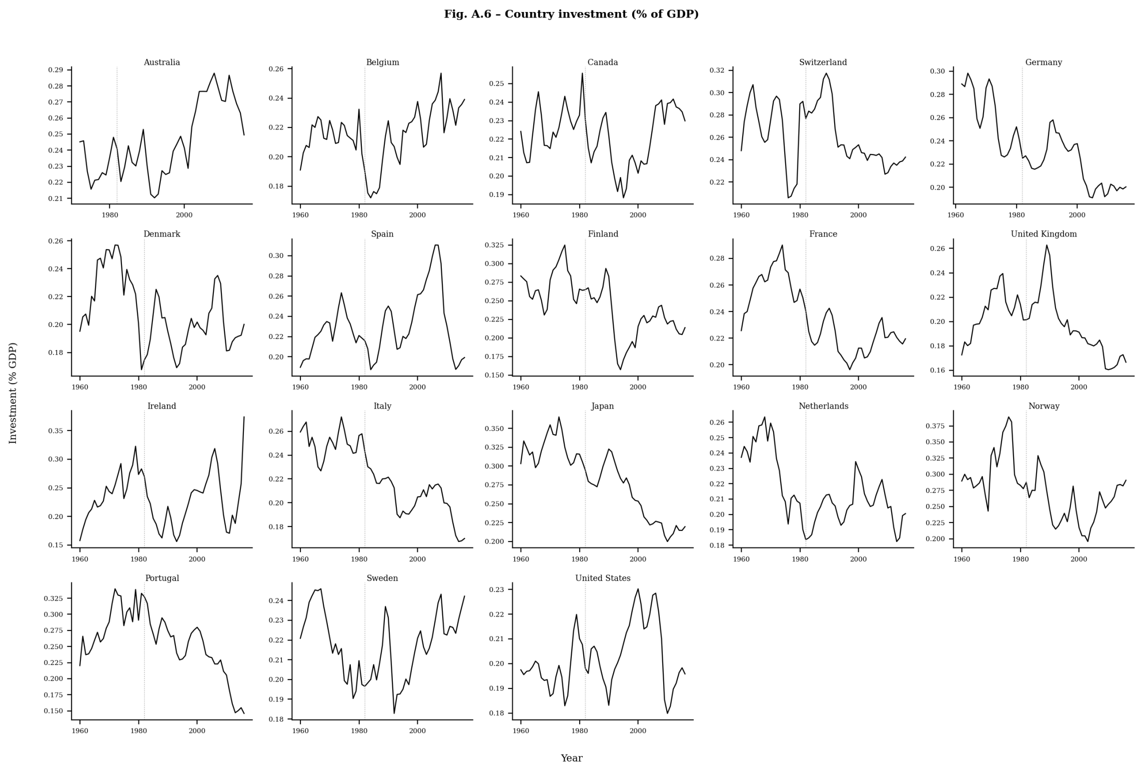

Figure 6: Fig. A6 – Investment ratio

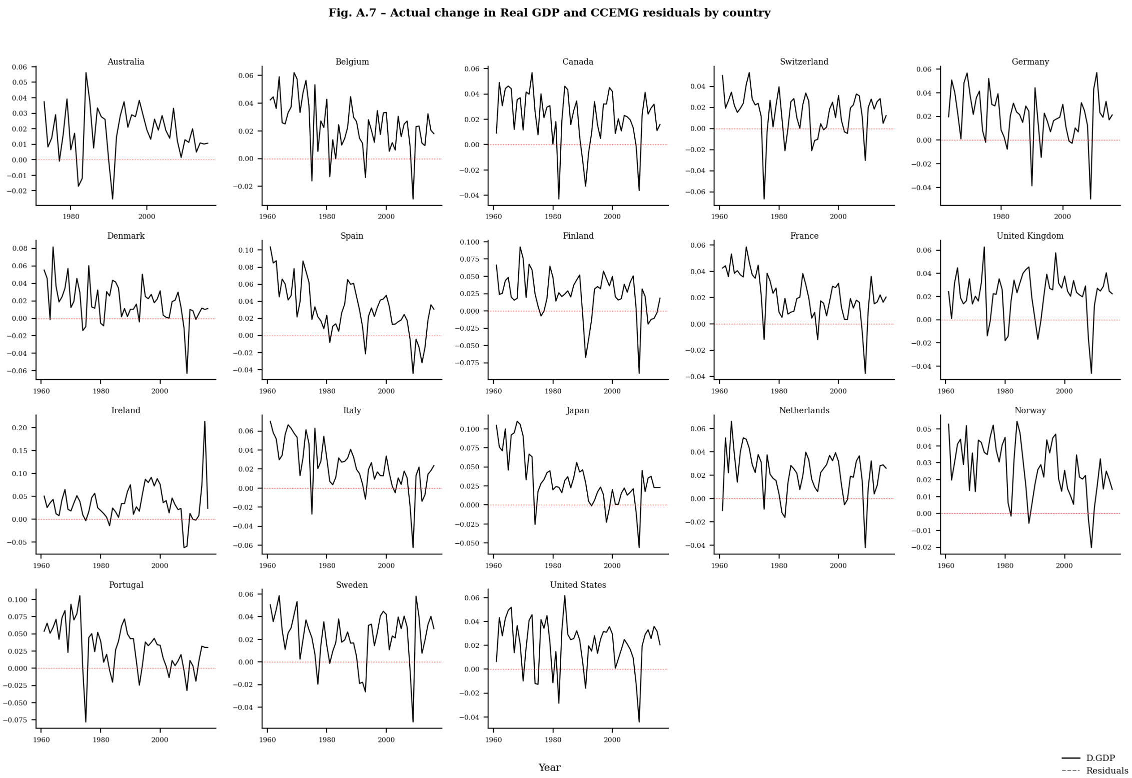

Figure 7: Fig. A7 – GDP growth and residuals

Optional and additional estimates

No additional exploratory models beyond the paper were added in this version. This section can be extended with subgroup estimates or alternative functional forms if needed.

To-do list for next panel data study

Tick	Items
	Verify data sources and update dates before any cleaning.
	Build a reproducible data pipeline with raw and processed datasets.
	Inspect missingness and define the panel sample selection rules.
	Compute between/within variance shares and classify variables.
	Produce descriptive plots for levels, within, between, and first differences.
	Run stationarity and cross-section dependence tests early.
	Estimate baseline FE/RE/FD models and check multicollinearity.
	Test dynamic specifications and instrument strength.
	Perform robustness checks (subsamples, outliers, policy periods).
	Document all code and results with a clear narrative.